and advanced beam control to dwell continuously on a single point of the moving target. Even minor structural vibrations from the host vehicle can throw the beam off course. Prolonged firing in a fixed direction triggers 'thermal blooming', an atmospheric condition wherein the beam heats the surrounding air, creating a thermal lens that deforms, defocuses, and significantly degrades the laser's energy density. Also, thermal blooming makes lasers significantly less effective against targets coming directly at them on a continuous path, known as 'down-the-throat' shots.

Atmospheric conditions. Atmospheric conditions like dust, smoke, fog, rain, and turbulence tend to reduce accuracy and effectiveness of lasers over long distances. This causes the beam to lose power and spread out, called beam blooming, before reaching the target. A combination of hardware modifications, wavelength selection, and beam modulation are used to counteract absorption, scattering, and turbulence.

Operational limitations. Unlike missiles that strike instantly, DEWs need a sustained 'dwell time' to destroy the target. This makes hitting fast-moving or manoeuvring threats difficult. Also, 'hardened' targets, using armour, ablative materials, or reflective coatings, partly deflect the energy reducing the impact. Rapidly spinning targets prevent lasers concentrating heat on a single spot, reducing the lethality. Adding more lasers as a solution is restricted by availability of space and power. Another limitation is the advanced software required for integration of DEWs into networks.

Legal constraints. DEWs are not covered under the international laws. Use of these weapons, specifically those designed to cause permanent blindness or unenhanced vision, is prohibited. International regulations are required to be framed to lay down the permissible use both during war and peacetime.

Human resource. Operating and maintaining DEWs requires expert manpower in fields such as AI, IT, optics, materials science, advanced software, and power electronics. Finding, training, and retaining such specialists is difficult.



Tonbo Imaging's WaveStrike, the first deployable Gen3 HPM system (Source: https://www.defencexp.com)

DEWs: Future growth and way ahead

To realise the full potential of the DEWs, their development needs to balance technological innovation, military usage, and legal framework without compromising on humanitarian values. Technological innovations are required in fields such as adaptive beam control, atmospheric compensation, precision beam alignment, AI-assisted tracking, sensor fusion (see Box 4), and resilient command-and-control networks. For space-based applications, advancement is necessary in high-yield orbital power systems, long range beam control, and hardware resilience against extreme radiation and thermal fluctuations. Particle beam weapons remain experimental. For them to become viable, research is necessary in areas such as compact high-energy accelerators and stable particle beam focusing mechanisms. The main humanitarian concern is related to the responsible use of DEWs as they inflict severe, agonising injuries that lack immediate external visibility, such as deep-tissue burns and permanent blindness. This challenges the established norms of proportionality and human suffering. Furthermore, their footprints being nearly invisible, verification and accountability of the source become very difficult. Deployment of DEWs in space, a globally shared domain preserved for peaceful exploration, violates space treaties.

To become self-sufficient in DEWs in the long term, India will need to develop indigenous production capabilities for critical electronic and electro-optical components. High-power semiconductor devices, power-electronics modules, high-speed processors, control electronics, RF and microwave components, solid-state amplifiers, precision optical and beam-steering components, high-speed photodetectors, radar and AESA modules, infrared and electro-optical sensors, thermal-management systems, and ruggedised embedded computing and communication systems are high priority equipment.

To operate effectively in the hyper-velocity environments, AI and ML integration is essential. These technologies will enable DEWs to detect, track, and engage multiple threats simultaneously in real time while automating beam control and decision-making. Ultimately, DEWs will be transformed into autonomous, rapid-fire shields capable of adapting to volatile conditions and sustaining precision under extreme stress. EFY